

Fever and Confusion: Meningitis, Encephalitis and Suppurative Brain Infections

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Learning Outcomes

At the end of this lecture you should be familiar with the:

- 1) Clinical approach in patients presenting with fever and confusion
- 2) Common differential diagnosis in patients presenting with fever and confusion
- 3) Common aetiologies, clinical presentation, investigations, and management of patients with meningitis, encephalitis and brain abscess

Differential Diagnosis of Patient with Fever and Confusion

- CNS infections
- Other non-CNS infections (+/- systemic involvement) with encephalopathy* (septic, metabolic, toxic)
- Sepsis in patients with primary psychiatric disorder
- Autoimmune encephalitis (e.g. anti-NMDA receptor encephalitis)
- Inflammatory disorders affecting the brain (e.g. vasculitis, cerebral lupus)
- Sepsis with nonconvulsive status epilepticus
- Thyroid storm

*Encephalopathy – disorder of the brain resulting in an altered mental state (impaired cognition, orientation and consciousness)

Approach to Patient with Fever and Confusion – History (I)

- Age
- Fever – onset, duration, pattern
- Headache – onset, duration, nature, location, severity
- Features of meningism (symptoms to suggest meningeal inflammation) – neck stiffness, photophobia
- Other clinical features of raised intracranial pressure e.g. vomiting
- Consciousness and mental state
- Other neurological symptoms
- Features of underlying infections (respiratory, ENT, dental, GI, urinary...)
- Progression of symptoms
- Previous treatment prior to presentation (** esp. recent use of antibiotics)
- Travel / contact history

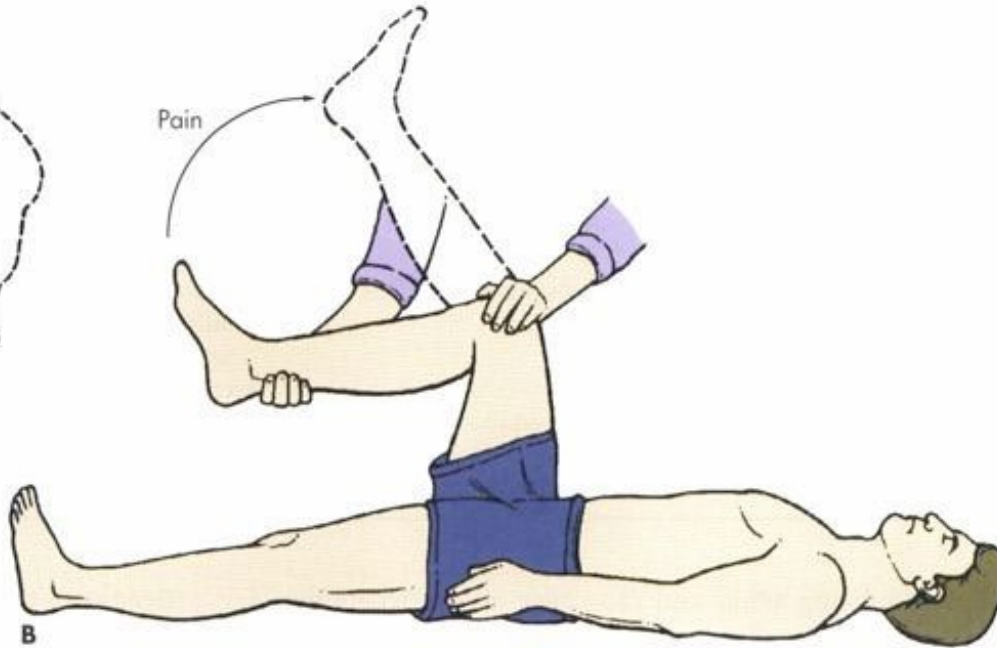
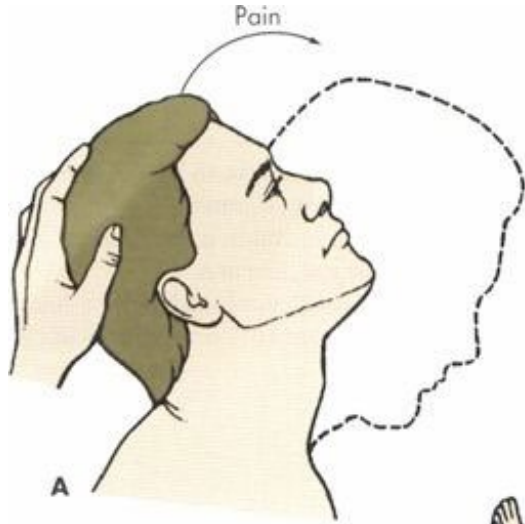
History (II)

- Past medical history:
 - Other chronic illness
 - History of tuberculosis
 - Immunocompromised state (e.g. diabetes, immunosuppressants, HIV)
 - Immunisation history
- Occupational history (e.g. *Streptococcus suis* meningitis in those who work with pigs / pork industry – meat processors, butchers)
- Drug history
 - Immunosuppressants (e.g. steroids, azathioprine, cyclosporine etc.)
 - Recent use of antibiotics
 - Other drug use, including over the counter and illicit drug use

Physical Examination

- Conscious level (Glasgow Coma Scale) and orientation
- Vital signs and temperature
- Signs of meningeal irritation – neck stiffness, Kernig's sign, Brudzinkin's sign
- Focal neurological signs
- Skin rash (e.g. morbilliform, petechial, purpuric in patient with meningococcal meningitis; vesicular rash in patients with herpes simplex or varicella-zoster virus infection)
- Other systems – dental, ENT (otitis media, sinusitis), respiratory, abdominal, cardiovascular (infective endocarditis)
- Fundi for papilloedema

Signs of Meningeal Irritation



A. Nuchal rigidity / neck stiffness

B. Kernig's sign – patient lying in supine position with hip and knee joint flexed to 90° . If passive extension of knee is limited by pain → positive Kernig's sign

C. Brudzinski's sign – upon flexion of the neck, there is involuntary flexing of the hips and knees → positive Brudzinski's sign

Investigations

- Blood tests
 - Complete blood count + differential count
 - Basic biochemistry (liver and renal function tests)
 - Inflammatory markers (ESR, CRP)
 - Blood culture
 - Clotting profile (? disseminated intravascular coagulopathy, preparation for lumbar puncture)
 - Blood gas
- Neuroimaging
 - CT brain (non-contrast)
 - To exclude mass lesion (e.g. brain abscess, malignancy) before lumbar puncture (risk of coning) esp. if patient has impaired consciousness or focal neurological signs
 - Should not delay treatment of presumed CNS infections
 - May reveal hypodensities (e.g. in encephalitis, ischaemic stroke complicating meningitis)

Investigations – Lumbar Puncture for CSF Sampling



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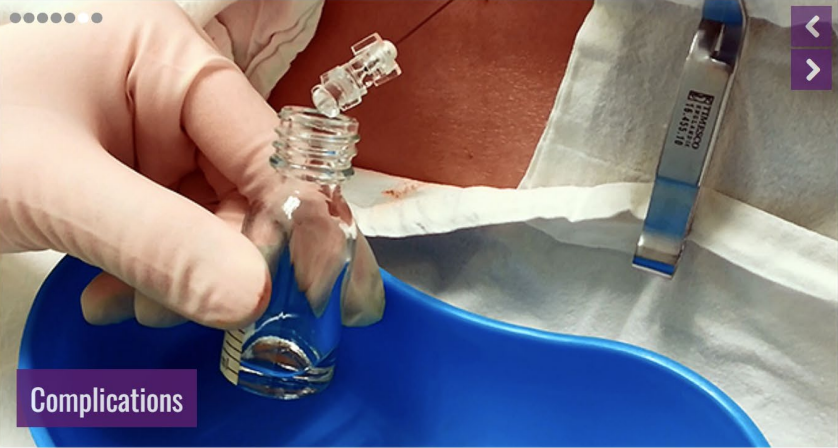
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Home / Neurology and Neurosurgery / How to Perform a Lumbar Puncture

How to Perform a Lumbar Puncture

Dr Gary Lau

In the following set of photos and videos, details of how a lumbar puncture is performed will be demonstrated. (Please note that medical students will not be expected to be perform lumbar punctures as part of their medical curriculum. These skills will be taught during Internship).



Complications

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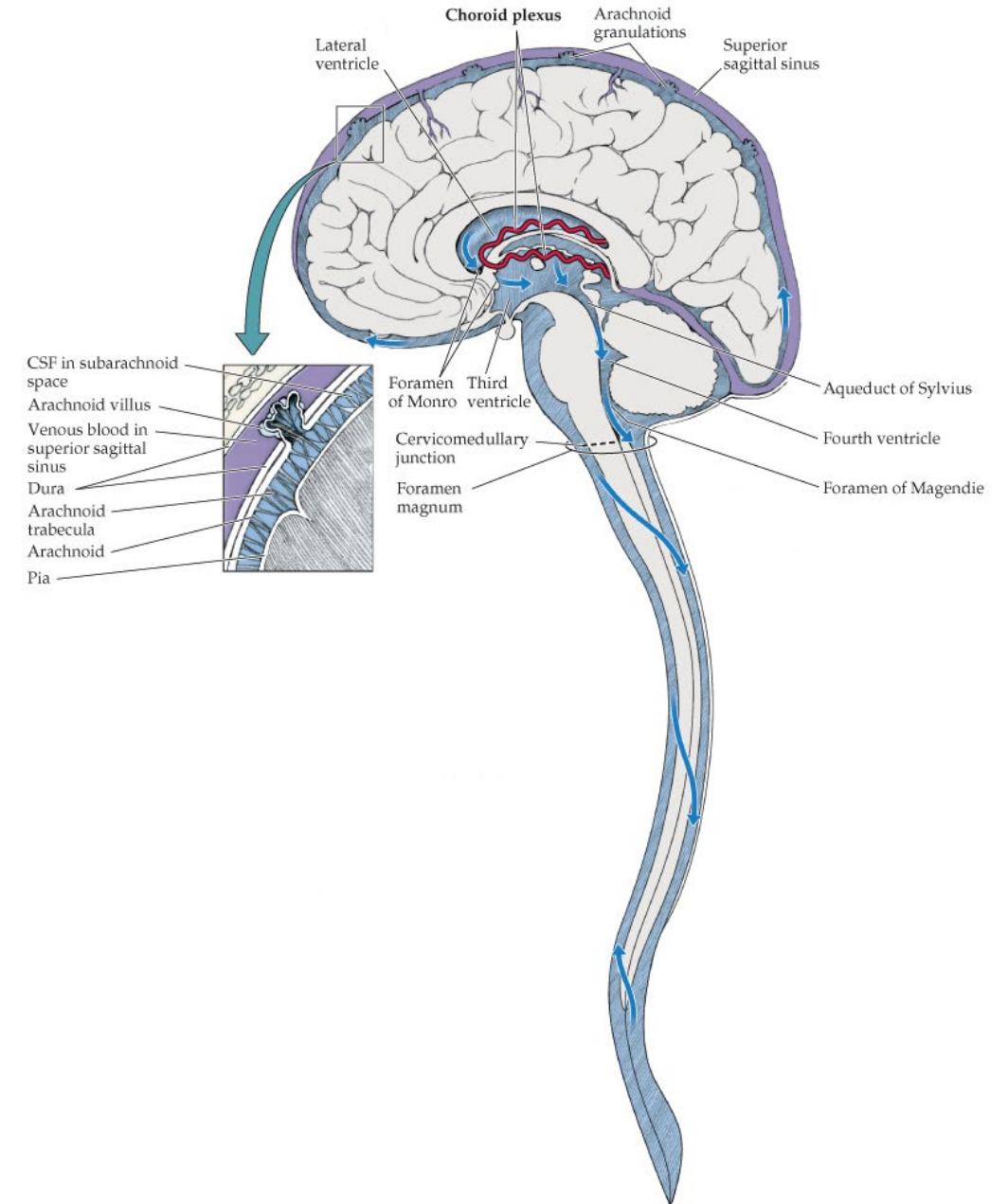
TAG CLOUD
Abducens nerveAtaxiaBrainstemCerebellar signsCerebellumCerebrospinal fluidCNS infectionCranial nerve



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Cerebrospinal Fluid

- Produced ~500ml/day
- Absorbed across the arachnoid villi by a valve-like mechanism
- Functions:
 - Mechanical support
 - Regulating ionic composition
 - Removal of metabolites



Lumbar Puncture - Indications

1. Diagnostic

- CNS infections, subarachnoid haemorrhage where CT noncontrast is nonrevealing (xanthochromia)
- Inflammatory conditions of the CNS (e.g. CNS vasculitis)
- Demyelinating disorders of the CNS (e.g. multiple sclerosis)
- Malignant disorders of the CNS

2. Therapeutic

- Analgesia (e.g. epidural anaesthesia)
- Intrathecal chemotherapy (e.g. haemic malignancies with CNS involvement)
- CSF removal (e.g. in patients with communicating hydrocephalus)



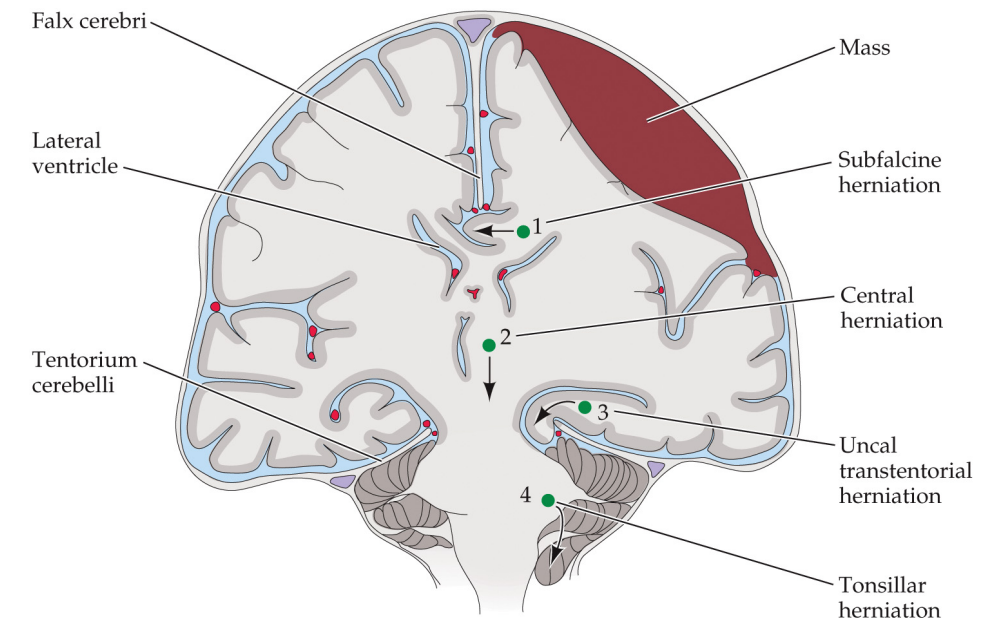
Lumbar Puncture – Contraindications & Complications

- Contraindications

- Intracranial lesion causing mass effect or significant cerebral oedema. LP may result in cerebral herniation
- Bleeding diathesis, e.g. severe coagulopathy (INR >1.4) or thrombocytopenia (platelet <50 x 10⁹/L). Spinal haematomas causing spinal cord compromise may result if bleeding diathesis not corrected
- Local congenital lesions
- Local suppuration

- Complications

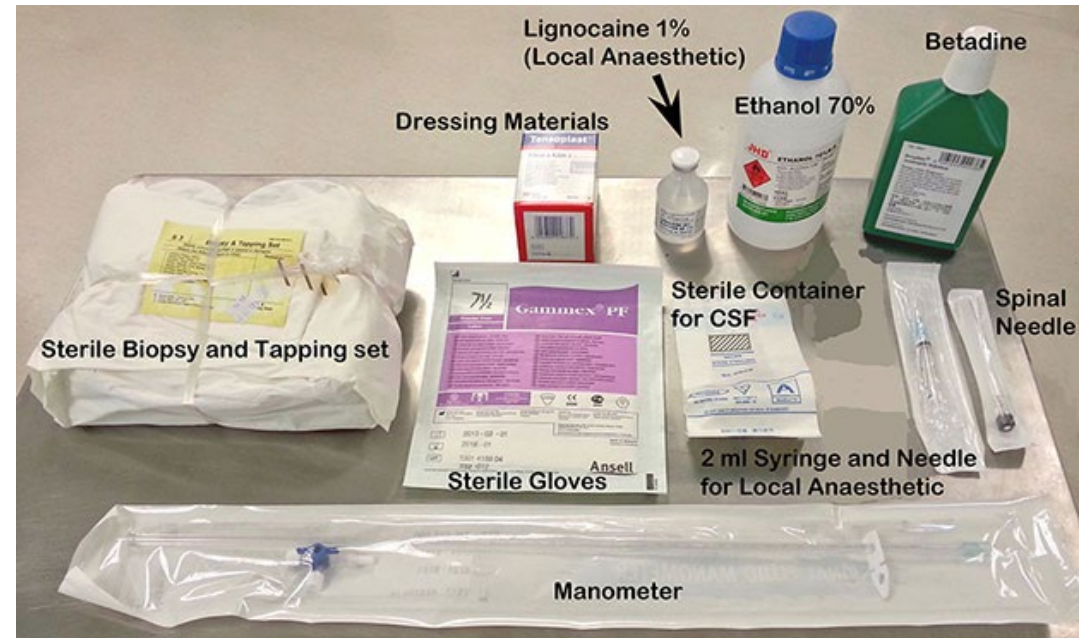
- **Headache**
- Radicular symptoms and low back pain
- Bleeding
- Infection
- Cerebral herniation



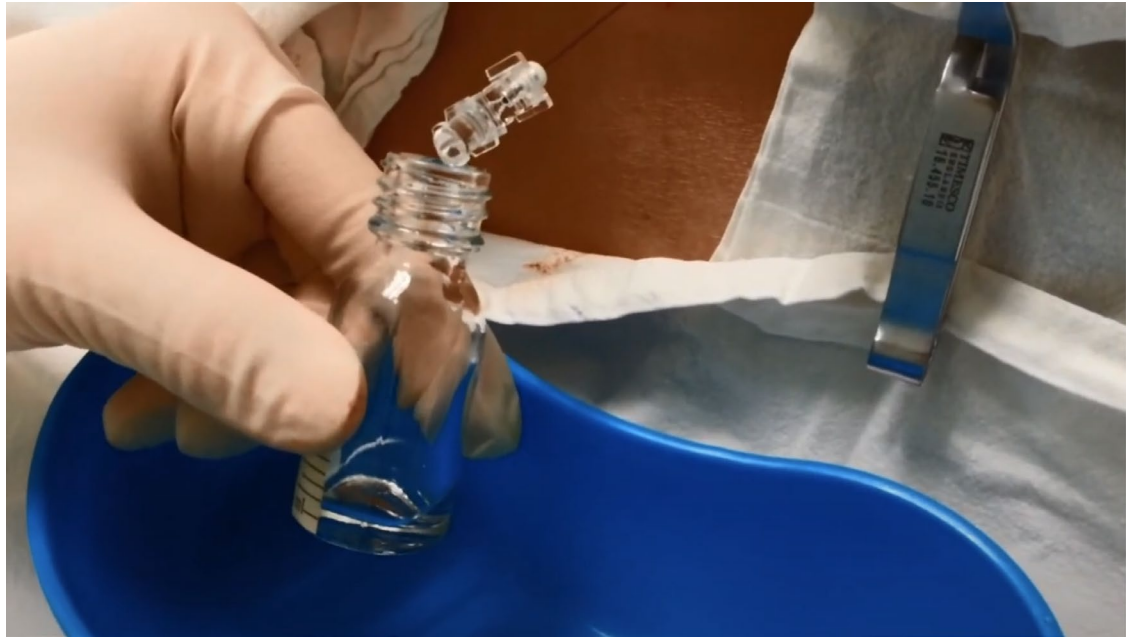
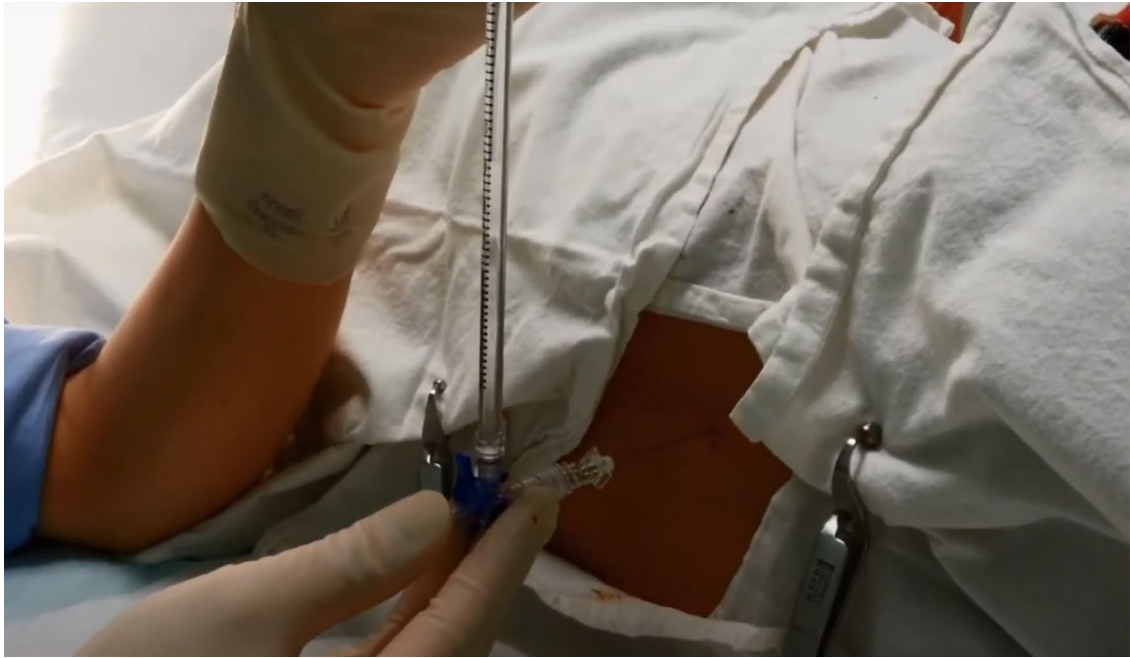
Lumbar Puncture - Preparation



Lateral recumbent position







Lumbar Puncture

- **Minimal 5ml should be collected
- Opening pressure (normal: 6-20cm H₂O, fluctuates with respiration)
- Colour (normal: colourless, turbid, pus, xanthochromia, blood-stained)
- Total cell count + differential count (normal <5cells/mL)
- Glucose (** + paired serum glucose **)
- Protein (0.15 – 0.45g/L)



Normal CSF



Xanthochromia

Xanthochromia – indicates presence of red cells in CSF. With time, the red cells breakdown, releasing haeme, which is subsequently degraded into the yellow-green pigment bilirubin causing characteristic yellow colour

Lumbar Puncture – Microbiology Workup

- Gram smear, bacterial culture
- Viral titre, culture and PCR for herpes simplex virus, enteroviruses and varicella zoster virus etc. [CSF + serum for Japanese B virus specific IgM antibodies if clinical suspicion]
- Ziehl-Neelson stains smear, mycobacterial culture, PCR for *Mycobacterium tuberculosis*
- Indian ink examination for cryptococcus, fungal culture (+/- cryptococcal Ag)
- (Multiplex PCR available to cover a range of common bacterial, viral and fungal organisms resulting in CNS infections)
- Other non-microbiological tests as indicated, e.g.
 - Oligoclonal bands if suspicion of demyelinating disorders
 - Cytology if suspicion of malignancy involving the CNS
 - Autoimmune markers if suspicion of autoimmune encephalitis

Typical CSF findings in CNS Infections

	Normal	Viral	Bacterial	Tuberculous	Fungal
Opening pressure	6-20cmH ₂ O	Normal/high	↑	↑	↑
Cells/mm ³	<5	50-1,000	1,000-5,000	50-300	20-500
Cell type (predominance)	Mononuclear	Mononuclear	Polymorphonuclear	Mononuclear	Mononuclear
CSF/plasma glucose ratio	>50%	Normal	↓ (<50% of blood glucose)	↓	↓
Protein (g/L)	<0.45	Normal / Mildly elevated	↑	↑	↑

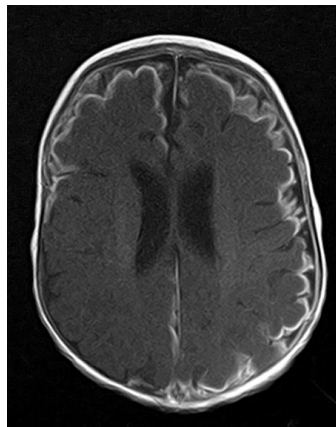
N.B. Partially-treated bacterial meningitis and malignant meningitis (e.g. due to lymphoma, leukaemia or carcinomatosis) may also have a CSF picture with lymphocytic predominant pleocytosis with low glucose and raised protein

Other Investigations as Indicated

- Chest X ray
- Nasopharyngeal aspirate for respiratory viruses
- Sputum for gram smear and bacterial culture; AFB smear and culture
- Blister fluid for viral studies
- Urine for routine microscopy and culture
- Blood for thyroid function tests
- Urine for toxicology

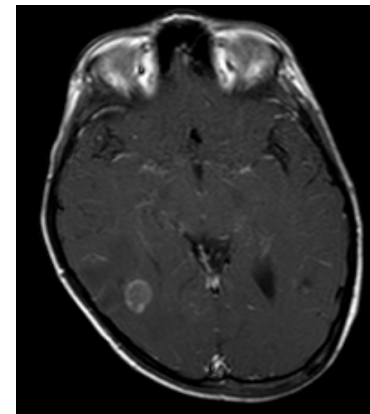
Other Investigations as Indicated

- CT brain with contrast - ? abscess, meningeal enhancement, tuberculomas
- MRI brain with contrast – ? meningeal enhancement, encephalitic changes, tuberculomas
- Electroencephalogram (EEG) – generalised slow waves (encephalopathic changes), periodic lateralising epileptiform discharges (e.g. in HSV encephalitis), other epileptiform activities
- Echocardiogram if suspect infective endocarditis, imaging of other adjacent structures (e.g. sinuses)



MRI brain with contrast showing meningeal enhancement

Case courtesy of Ammar Haouimi, Radiopaedia.org, rID: 97444



MRI brain with contrast showing tuberculoma

Case courtesy of Ahmed Abdrabou, Radiopaedia.org, rID: 41152

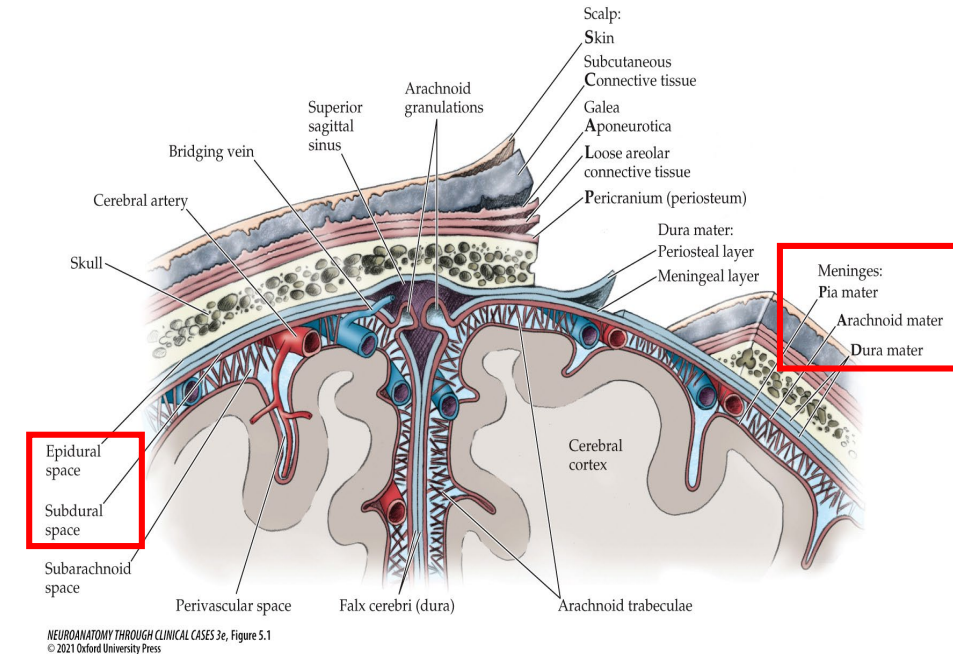
Management

- Empirical management without delay
 - Initially with broad spectrum antibiotics that penetrate the blood brain barrier
 - E.g. high dose **IV ceftriaxone** / meropenem
 - **Aciclovir** for coverage of viruses (herpes simplex virus and varicella-zoster virus)
 - N.B. IV vancomycin if suspected methicillin resistant *Staphylococcus aureus*
 - Antibiotics streamlined according to microbiological workup results
- Neurosurgical consultation if hydrocephalus (external ventricular drainage / shunting)



General Points on CNS Infections (I)

- Can involve different structures:
 - Meninges: meningitis
 - Brain parenchyma: encephalitis
 - Spinal cord: myelitis
 - Pus within the brain parenchyma: brain abscess
 - Pus outside the brain parenchyma: epidural or subdural empyema
- Potential micro-organisms:
 - Bacteria
 - Viruses
 - Fungi
 - Protozoa (rare)
- Can be acute (hours – days), subacute (days - weeks) and chronic (symptoms >4 weeks)



General Points on CNS Infections (II)

- Common
- High mortality and morbidity if treatment delayed
- Quick septic workup including blood culture before start of antibiotics
- Early lumbar puncture if no contraindications
- Initial high-dose broad spectrum parenteral antibiotics that penetrates the blood brain barrier, subsequent streamlining of antibiotics with availability of microbiological results
- Regular monitoring of neurological status and vital signs
- Look for the primary infective focus and also complications
- Close liaison with microbiologist, neurologist and neurosurgeon

Meningitis

- Inflammation of the leptomeninges: infection (bacterial, viral, fungal), neoplastic infiltration, (irritation by drugs resulting in drug-induced aseptic meningitis)
- Acute meningitis – bacteria or viruses
- Subacute / chronic meningitis – mycobacteria, fungi or atypical bacteria
- Pathophysiology
 1. **Direct spread** from nearby structures, e.g. infections of the respiratory tract, ear, sinus, dental source, skull fracture etc.
 2. **Haematogenous spread** from a distant septic foci

Bacterial Meningitis (I) – Pathophysiology

Infection stimulates an immune response → pia-arachnoid membrane infiltrated with inflammatory cells → pus forms which can organise to form **adhesions**

→ obstruct free flow of CSF → **hydrocephalus**

→ **damage cranial nerves** at base of brain, can result in hearing loss

→ obliterative endarteritis of leptomeningeal arteries passing through the meningeal exudate can result in secondary **cerebral infarction**

Bacterial Meningitis (II) – Common Organisms

Vary according to age of onset and also geographical region

18 – 50 years	<i>Streptococcus pneumoniae</i> , <i>Neisseria meningitidis</i>
>50 years	+ Gram negative bacilli (e.g. <i>Klebsiella</i> sp., <i>Escherichia</i> sp., <i>Salmonella</i> sp.)
Immunocompromised host	+ <i>Listeria monocytogenes</i> , <i>Pseudomonas aeruginosa</i>
Occupational / exposure to pigs	<i>Streptococcus suis</i>
Exposure to contaminated fresh water	<i>Nagleria fowleri</i> & other free-living amoeba
Head trauma, intrathecal injection and post-neurosurgery	<i>Staphylococcus aureus</i> , <i>Staphylococcus epidermidis</i> , aerobic gram-negative bacilli

Bacterial Meningitis (III)

- Headache, fever and meningeal irritation (neck stiffness, Kernig's sign, Brudzinksi's sign..)
- Severe meningitis: impaired consciousness (encephalopathy, cerebral oedema), focal neurological signs
- Complications of meningococcal sepsis
 - Meningitis
 - Rash (morbilliform, petechial, purpuric)
 - Septic shock
 - Disseminated intravascular coagulation
 - Renal failure
 - Peripheral gangrene



Bacterial Meningitis (IV) - Management

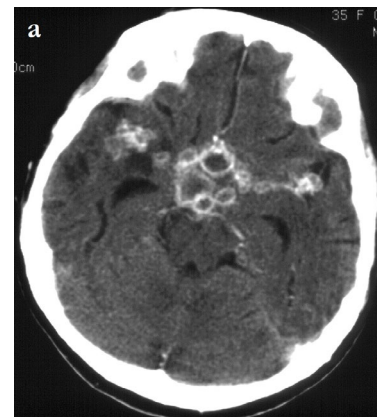
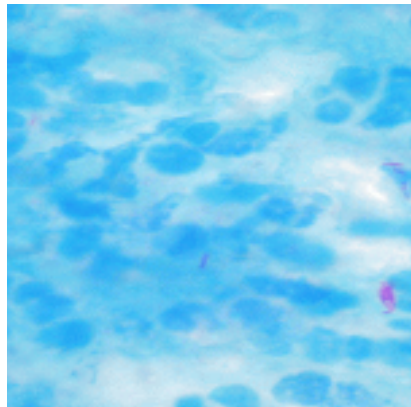
- High mortality rate of 80% if untreated
- Empirical high-dose antibiotics that penetrates the blood brain barrier, regimen subsequently guided by culture results
- Benefit of adjunctive dexamethasone remains inconclusive
- Close contacts of patients suffering from *Haemophilus influenza type B* and *Neisseria meningitidis* require chemoprophylaxis (e.g. rifampicin or ciprofloxacin)

Tuberculous Meningitis (I)

- Very common cause of meningitis in Hong Kong
- Pathophysiology:
 - Rupture of a superficial infective granuloma on the pia mater into the subarachnoid space
 - May have concomitant pulmonary / systemic disease
- Clinical presentation
 - Onset slower than bacterial meningitis
 - Fever and headache for several days to weeks

Tuberculous Meningitis (II) - Investigations

- Lumbar puncture: low sensitivity of Ziehl-Neelsen smear (<15%), CSF culture (<50%); TB PCR (~50-90%)
- Neuroimaging (CT/MRI):
 - MRI: may show hydrocephalus, meningeal enhancement, intracranial tuberculomas, cerebral infarct
- Chest radiograph, sputum x AFB smear and culture

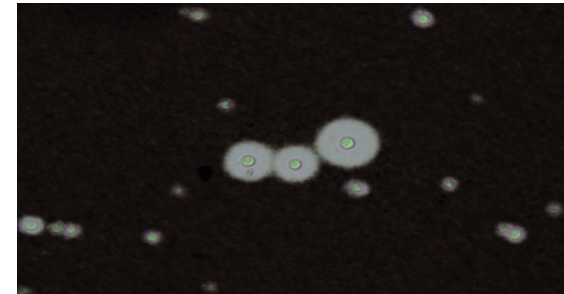


Tuberculous Meningitis (III) - Management

- High mortality and morbidity amongst survivors
- **Anti-TB treatment** should be started if suspicious of TB meningitis (even if smear and PCR negative as limited sensitivity)
 - Isoniazid, rifampicin, pyrazinamide, ethambutol
 - Duration of treatment **at least 9 to 12 months**, often extended to 18 months if immunocompromised (e.g. with HIV) due to higher risk of relapse and developing drug-resistance TB
- Concomitant use of **dexamethasone** (tapering dose, given parenterally initially) – reduces risk of death and disability

Cryptococcal Meningitis

- Presentation
 - **Subacute onset** of headache and cognitive impairment
 - 50% have **underlying immunodeficiency disorder** (e.g. HIV, receiving immunosuppressants such as steroids, history of transplantation etc.)
- Investigations
 - Lumbar puncture: lymphocytic pleocytosis, high protein, low CSF to serum glucose, positive Indian ink exam (50%), positive CSF cryptococcal antigen (85%), fungal culture (90%)
 - Serum cryptococcal antigen
- Management
 - **Antifungal treatment**
 - **IV amphotericin B + oral flucytosine** (induction) followed by prolonged course of **oral fluconazole** (maintenance), total duration usually weeks to months



'Negative' stain corresponds to the halo of capsular material which encloses the budding yeast

Viral Meningitis

- Presentation
 - **Benign and self-limiting illness**
 - Much less serious than bacterial meningitis unless there is associated encephalitis
- Common viruses: **enteroviruses** (echo, Coxsackie, polio viruses), herpes simplex types 1 and 2, varicella zoster virus, cytomegalovirus, Epstein-Barr virus, HIV,
- Management
 - Symptomatic treatment
 - Usually self-limiting with complete recovery within days without specific therapy

Complications of Meningitis

- Meningeal adhesions → raised intracranial pressure, obstructive hydrocephalus, cranial nerve palsies
- Arteritis / thrombophlebitis → cerebral infarction
- Seizures and epilepsy
- Local spread of infection (cerebritis, cerebral abscess, subdural effusion / empyema)
- Intellectual impairment, cerebral palsy
- Syndrome of inappropriate secretion of anti-diuretic hormone (SIADH)
- Disseminated intravascular coagulation
- Septic shock

Encephalitis

- Inflammation of the brain parenchyma, mostly **viral**
- Usually with meningitis
- Aetiological agents:
 - **Herpes viruses** (herpes simplex and varicella-zoster viruses) – **HSV-1** most common
 - **Enteroviruses** (Coxsackie virus, echo virus, poliovirus)
 - Influenza virus
 - Adenovirus
 - Cytomegalovirus
 - Epstein-Barr virus
 - HIV
 - Flaviviruses (e.g. Japanese encephalitis virus)
 - Rabies virus

Viral Encephalitis (I) - Pathophysiology

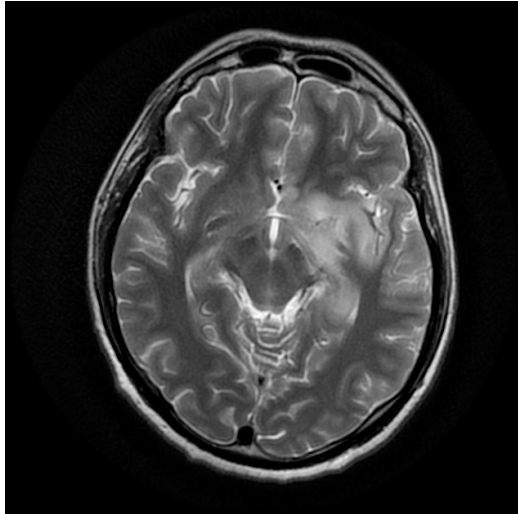
- Inflammatory response that involves the cortex, white matter, basal ganglia and brainstem
- Distribution of lesions varies with the type of virus
 - Herpes simplex encephalitis – **frontal and temporal lobes**
 - Cytomegalovirus – areas adjacent to the ventricles (ventriculitis)
 - Japanese encephalitis virus – thalamus
- Inclusion bodies may be present in the neurons and glial cells and there is an infiltration of polymorphonuclear cells in the perivascular space
- Neuronal degeneration and diffuse glial proliferation, often with cerebral oedema

Viral Encephalitis (II) - Presentation

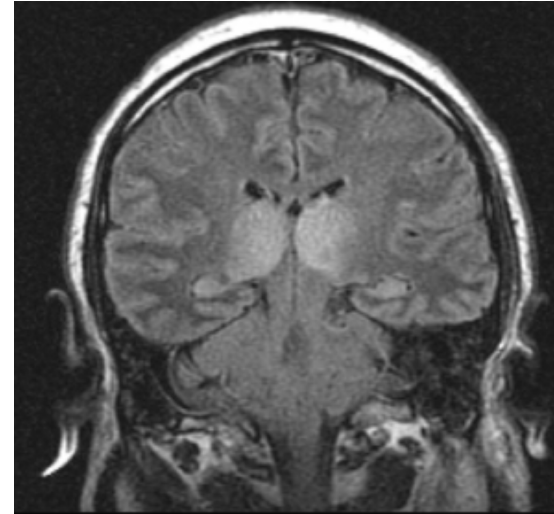
- Fever
- Headache
- Focal neurological signs – dysphasia, hemiplegia, visual field defects etc.
- Seizures
- Confusion
- Altered level of consciousness – mild lethargy to deep coma

Viral Encephalitis (III) – Investigations

- Neuroimaging (CT / MRI)
 - Hypodensity (CT) / hyperintensity (MRI) in frontotemporal lobes (HSV)
 - Hyperintensity (MRI) of bilateral thalamus in Japanese encephalitis



MRI brain (T2, axial cut): hyperintensity of the left medial temporal lobe in patient with HSV-1 encephalitis



MRI brain (coronal cut): hyperintensity of the bilateral thalamus in patient with Japanese B encephalitis

Viral Encephalitis (IV) – Investigations & Management

- Investigations

- Lumbar puncture
 - Lymphocytic predominant pleocytosis (neutrophils may predominate in early stages)
 - Normal glucose, mild elevation in protein
 - Abnormal viral culture / PCR
- EEG – periodic slow-wave activity / lateralisation epileptiform discharges involving the frontotemporal lobes in HSV encephalitis

- Treatment

- **Aciclovir** 10mg/kg IV three times per day for 2-3 weeks for HSV encephalitis
 - Should be started in all patients with suspected viral encephalitis
 - Significantly reduced mortality of HSV encephalitis
- Antiseizure medications may be needed

Parenchymal Bacterial Infections – Cerebral Abscess (I)

- Pathophysiology
 - Bacteria may enter the brain substance by:
 - **Penetrating trauma**
 - **Direct spread** from paranasal sinuses or the middle ear
 - **Haematogenous spread** may lead to multiple abscesses
 - *Untreated congenital heart disease is a risk factor (right to left shunting can result in bacteria to directly affect the brain; increased risk of developing infective endocarditis; immunocompromised state)*
 - Site of abscess and causative organism related to the source of infection
 - Initial infection leads to local suppuration followed by loculation of pus within a surrounding wall of gliosis

Cerebral Abscess (II) - Presentation

- Presentation
 - Can present over days to weeks
 - Headache
 - Changes in mental status
 - Focal neurological deficits
 - Seizures
 - Raised intracranial pressure

Cerebral Abscess (III) – Aetiological Agents

- Depends on underlying predisposition, e.g.
 - Otitis media, mastoiditis – Streptococci, *Bacteroides spp.*
 - Sinusitis – Streptococci, *Bacteroides*, *Staphylococcus aureus*, *Haemophilus spp.*
 - Dental infection – Mixed anaerobes with *Fusobacterium*, *Bacteroides spp.*, streptococci
 - Penetrating trauma / post-surgery – *Staphylococcus aureus*, streptococci
 - Infective endocarditis – *Staphylococcus aureus*, streptococci
 - HIV infection – *Toxoplasma gondii*, *Nocardia spp.*, *Cryptococcus neoformans*

Cerebral Abscess (IV) - Investigations

- CT brain: single or multiple hypodense lesions, ring enhancement with contrast, surrounding cerebral oedema
- Lumbar puncture potentially hazardous
- Stereotactic CT-guided aspiration or incision and drainage are important to aid microbiological diagnosis



Non-contrast CT brain: round intra-axial lesion at the left temporal lobe with vasogenic oedema



Contrast CT brain: ring enhancement of the left temporal lobe lesion

Cerebral Abscess (V) – Management & Prognosis

- Broad spectrum antimicrobial therapy with **coverage of both aerobic and anaerobic bacteria**
 - E.g. **Ceftriaxone IV + Metronidazole IV**
- Likely source of infection can help guide antibiotic choice
- Neurosurgical patients – consider adding IV vancomycin
- **Surgical drainage** by Burr-hole aspiration may be necessary
- Anti-seizure medications if develop seizure
- Intravenous mannitol / steroids if significant cerebral oedema
- 10-20% mortality

Autoimmune Encephalitis (I)

- Autoimmune encephalitis is a differential diagnosis of patients presenting with fever and confusion
- Rare
- May be associated with **underlying malignancies** and infections that have triggered an immune response against the brain parenchyma
- Increasing number of antibodies being recognised - most common being **anti-NMDA (N-methyl-D-aspartate) receptor encephalitis**
 - Most commonly affects **young women** and children
 - Associated with **ovarian teratomas**
 - Clinical presentation:
 - Viral-like prodrome
 - Neuropsychiatric symptoms (altered cognition, behavioural changes – psychosis, aggression, catatonia..)
 - Abnormal movements (orofacial, limb or trunk dyskinesias)
 - Seizures
 - Altered consciousness
 - Autonomic dysfunction

Autoimmune Encephalitis (II)

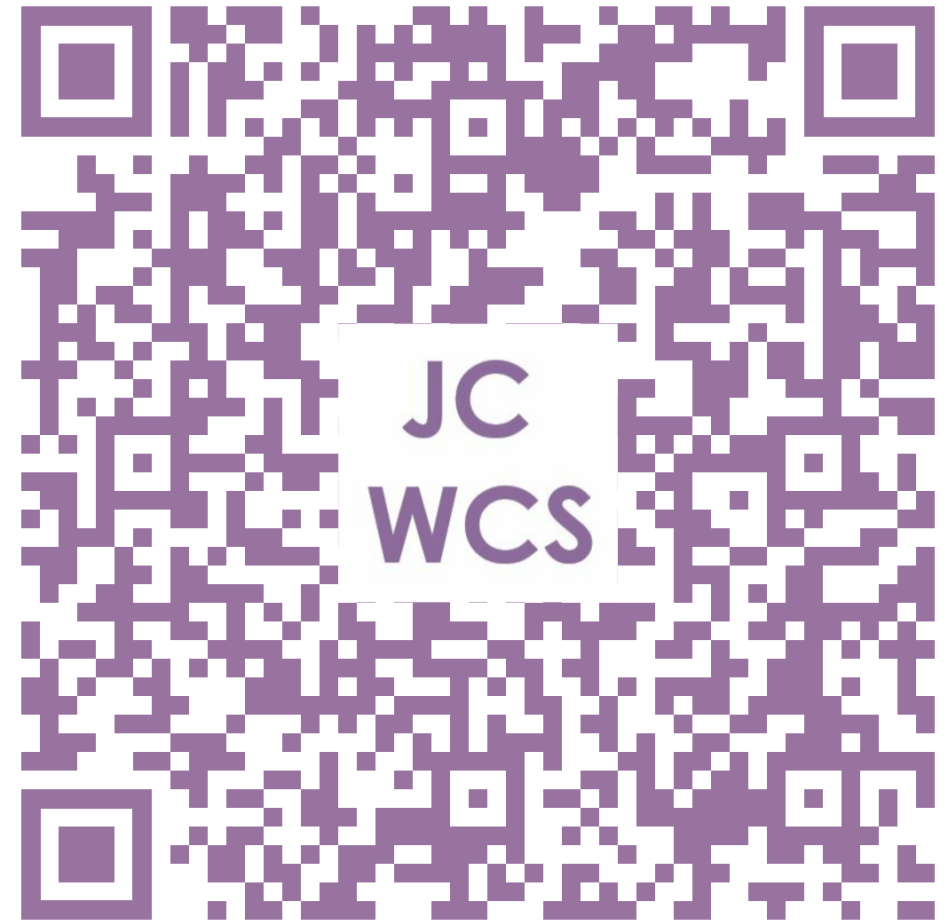
- Initial investigation and management approach similar to that of CNS infections
- Infective aetiologies must be excluded
- **Blood and CSF for autoimmune antibodies** (e.g. anti-NMDA receptor antibodies) if clinical suspicion
- Management
 - **Close monitoring** as patients may develop refractory seizures, autonomic instability
 - **Immunotherapy**
 - Acute stage – high dose intravenous prednisolone, intravenous immunoglobulin, plasmapheresis
 - Subsequent treatment with oral corticosteroids / steroid sparing agents
 - Anti-seizure medications if seizures
 - Supportive care

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)